The Journal of Open Source Software

measles: Tools for Simulating and Investigating Measles Outbreaks with R

George G Vega Yon[1], **Damon Toth**[1], **Jake Wagoner**[1], and **Olivia Banks**[1]

1 The University of Utah, United States

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Sehrish Kanwal

Reviewers:

- @JDRomano2
- @welch16

Submitted: 23 February 2026
Published: unpublished

Summary

The `measles` package implements agent-based models for simulating measles outbreaks in school settings and community populations in the R programming language (R Core Team, 2025). Developed through collaborations with public health partners during the 2025 measles outbreak in the United States, it depends on `epiworldR` (Meyer et al., 2025; Meyer & Vega Yon, 2023) for its core simulation engine and provides three specialized measles models that users can apply to simulate outbreaks and evaluate the potential effects of different intervention strategies.

Statement of need

The ongoing U.S. measles outbreak, which began in Texas in 2025, has resulted in thousands of cases across the United States. In parallel with the rise of anti-vaccination campaigns, diseases such as measles—previously considered eliminated in many settings—have re-emerged as significant public health concerns. Measles is a highly transmissible yet preventable disease, but declining immunization coverage over the past decade has reduced population immunity below the levels required to prevent outbreaks, which is commonly estimated at approximately 93% based on a basic reproductive number of approximately 15 (Guerra et al., 2017).

Although agent-based models of measles exist, many are not readily accessible to non-expert users or designed for routine use by public health practitioners. During the 2025 outbreak, public health partners explicitly identified the need for a user-friendly, well-documented measles modeling tool that could be used to support decision-making in real time (*Insight Net Annual Meeting 2025*, n.d.).

The models implemented in the `measles` R package were developed and refined through extensive use by public health officials during outbreak response efforts. As a result, the package provides field-tested, efficient, and well-documented agent-based models that bring advanced modeling capabilities closer to practitioners. By lowering technical barriers, the `measles` package enables epidemiologists and public health officials to explore outbreak dynamics and evaluate intervention strategies using a tool grounded in real-world public health practice.

State of the field

Measles transmission has been studied using a range of modeling approaches, including agent-based models, compartmental models, and interactive simulators aimed at public health communication and planning. Existing tools differ substantially in their intended audience, accessibility, and extensibility. Taken together, these tools do not provide a measles-specific agent-based framework that is both extensible and readily usable within the R ecosystem.

Several widely used measles simulators are designed primarily as interactive, practitioner-facing tools. The Framework for Reconstructing Epidemiological Dynamics (FRED) provides an agent-based modeling platform built on synthetic populations with realistic household and school structures and includes a U.S. measles simulator for scenario exploration (Grefenstette et al., 2013). The Centers for Disease Control and Prevention (CDC) offers an interactive measles outbreak simulator that supports exploration of outbreak trajectories under different intervention strategies using a stochastic discrete-time compartmental model, but it is primarily intended for communication rather than extensible research workflows (CDC, 2026). Similarly, the epiENGAGE measles simulator focuses on accessibility and visualization for school-based outbreaks, rather than reusable modeling components, without implementing interventions other than vaccination (*epiENGAGE Measles Outbreak Simulator v-2.6.0*, n.d.).

Within the R ecosystem, measles modeling is primarily achieved through general-purpose infectious disease modeling and inference frameworks rather than measles-specific outbreak simulators. Packages such as `pomp` and `panelPomp` include canonical measles examples for methodological development and inference but typically require substantial modeling expertise and project-specific implementation (Bretó et al., 2025; He et al., 2009).

Overall, existing tools tend to emphasize either accessibility with limited extensibility or flexibility with high technical barriers. This gap highlights the need for a reusable, agent-based measles modeling package that is accessible to applied epidemiologists and public health practitioners working within the R ecosystem.

Software design

The `measles` package builds upon the `epiworld` C++ library, which provides a general framework for fast agent-based models. This design enables efficient simulation while retaining the flexibility needed for applied outbreak modeling.

All models in `measles` are discrete-time agent-based models that differ in how contacts between agents are represented. The package includes three approaches: a single-school model assuming perfect mixing; a community-level model that uses a mixing matrix to represent within- and between-group interactions; and an extended model that incorporates risk-dependent quarantine durations.

Each model implements the full disease progression of measles, including susceptible, latent, prodromal, rash, and recovered states, as well as hospitalization and quarantine processes. The API allows users to modify key components such as contact rates, individual susceptibility, index cases, initial conditions, and quarantine policies.

An extended example in the form of a vignette simulates an outbreak in the Utah-Arizona border region (encompassing Hildale, Utah and Colorado City, Arizona—collectively known as Short Creek—an area with low vaccination coverage). The simulation integrates school-level vaccination data from both the Utah Department of Health and Human Services and publicly available Arizona school data, population age structure from the U.S. Census for both states, and age-specific mixing patterns from the POLYMOD survey.

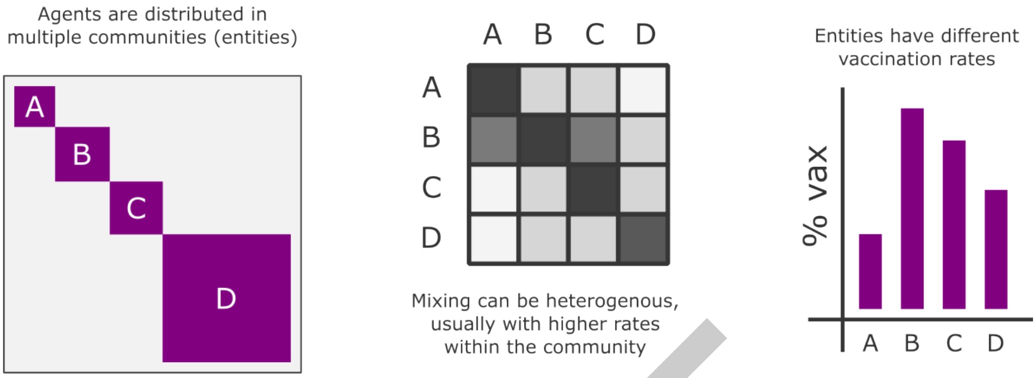

Figure 1: Mixing model architecture in the `measles` package showing entities (e.g., age groups, schools) and their interactions via a mixing matrix. Each entity can have different vaccination rates.

The package and its documentation (vignettes, API references, and help files) are readily available in the Comprehensive R Archive Network (CRAN) and on GitHub at https://github.com/UofUEpiBio/measles, where development takes place. Community guidelines, including a code of conduct, are available in the repository.

Research impact statement

The `measles` R package is the result of intensive collaboration between the University of Utah and the Utah Department of Health and Human Services (DHHS). Using the functions implemented in this package, and with funding from the U.S. Centers for Disease Control and Prevention, the University of Utah developed a Shiny application that has been used extensively by DHHS and other public health departments in the United States (Vega Yon et al., 2026).

By mid-2025, DHHS used the school-based models to generate individualized outbreak simulations for every school in the state of Utah. These results were communicated directly to school districts to support preparedness and response planning.

The `measles` package continues to be used through active collaborations with public health partners, including the states of Utah, Minnesota, and Washington, where the models are being used to generate measles outbreak simulations to help inform public health policy about the benefits of increased vaccination rates and quarantine and isolation for controlling outbreaks. Additional measles-related efforts, along with new and forthcoming tools from our team, are available at https://github.com/EpiForeSITE/measles.

AI usage disclosure

Generative AI tools were used in the development of this work. GitHub Copilot was used to assist with portions of the software implementation. ChatGPT was used to assist with drafting and editing parts of this manuscript. All AI-assisted content was reviewed and validated by the authors. The full conversation for the ChatGPT-assisted component is available at this link.

Acknowledgements

This work was supported by the Centers for Disease Control and Prevention InsightNet award number CDC-RFA-FT-23-0069.

References

Bretó, C., Wheeler, J., King, A. A., & Ionides, E. L. (2025). panelPomp: Analysis of panel data via partially observed markov processes in r. *The R Journal*, *17*, 180–199. https://doi.org/10.32614/RJ-2025-009

CDC. (2026). Measles Outbreak Simulator. In *CFA: Modeling and Forecasting*. https://www.cdc.gov/cfa-modeling-and-forecasting/measles-outbreak-simulator/index.html.

epiENGAGE Measles Outbreak Simulator v-2.6.0. (n.d.). https://epiengage-measles.tacc.utexas.edu/.

Grefenstette, J. J., Brown, S. T., Rosenfeld, R., DePasse, J., Stone, N. T., Cooley, P. C., Wheaton, W. D., Fyshe, A., Galloway, D. D., Sriram, A., Guclu, H., Abraham, T., & Burke, D. S. (2013). FRED (A Framework for Reconstructing Epidemic Dynamics): An open-source software system for modeling infectious diseases and control strategies using census-based populations. *BMC Public Health*, *13*(1), 940. https://doi.org/10.1186/1471-2458-13-940

Guerra, F. M., Bolotin, S., Lim, G., Heffernan, J., Deeks, S. L., Li, Y., & Crowcroft, N. S. (2017). The basic reproduction number (R0) of measles: A systematic review. *The Lancet Infectious Diseases*, *17*(12), e420–e428. https://doi.org/10.1016/S1473-3099(17)30307-9

He, D., Ionides, E. L., & King, A. A. (2009). Plug-and-play inference for disease dynamics: Measles in large and small populations as a case study. *Journal of The Royal Society Interface*, *7*(43), 271–283. https://doi.org/10.1098/rsif.2009.0151

Insight Net Annual Meeting 2025. (n.d.). https://www.eventleaf.com/insight-net-annual-meeting-2025#ss.

Meyer, D., Pulsipher, A., & Vega Yon, G. (2025). *epiworldR: Fast Agent-Based Epi Models*. https://github.com/UofUEpiBio/epiworldR

Meyer, D., & Vega Yon, G. (2023). epiworldR: Fast agent-based epi models. *The Journal of Open Source Software*, *8*(90). https://doi.org/10.21105/joss.05781

R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Vega Yon, G. G., Thornton, D., Redd, A., Pulsipher, A., Toth, D., Dorsan, E., Rennert, L., Johnson, K., White, L. F., Gruninger, R., Nolen, L. D., & Samore, M. H. (2026). *Practical Guidelines and Reflections on Building Public Health Software: A Measles Case Study (Preprint)*. JMIR Public Health and Surveillance. https://doi.org/10.2196/preprints.94240